

A ground-based autonomous rover system that collects detailed multi-modal sensor data in shrubland regions



Problem

Land managers and researchers lack reliable, ground-level data for assessing vegetation health and land conditions in shrubland environments.

No Ground-Level Data

(Susan Prichard · UW FLAME Lab)

Aerial & satellite sensing misses fine-scale variation in vegetation and fuel conditions critical for fire risk assessment.

Existing Sensors Can't Reach the Ground

UAVs blocked by canopy, wind, terrain & no-fly zones. Satellite data averages across too large an area to be actionable.

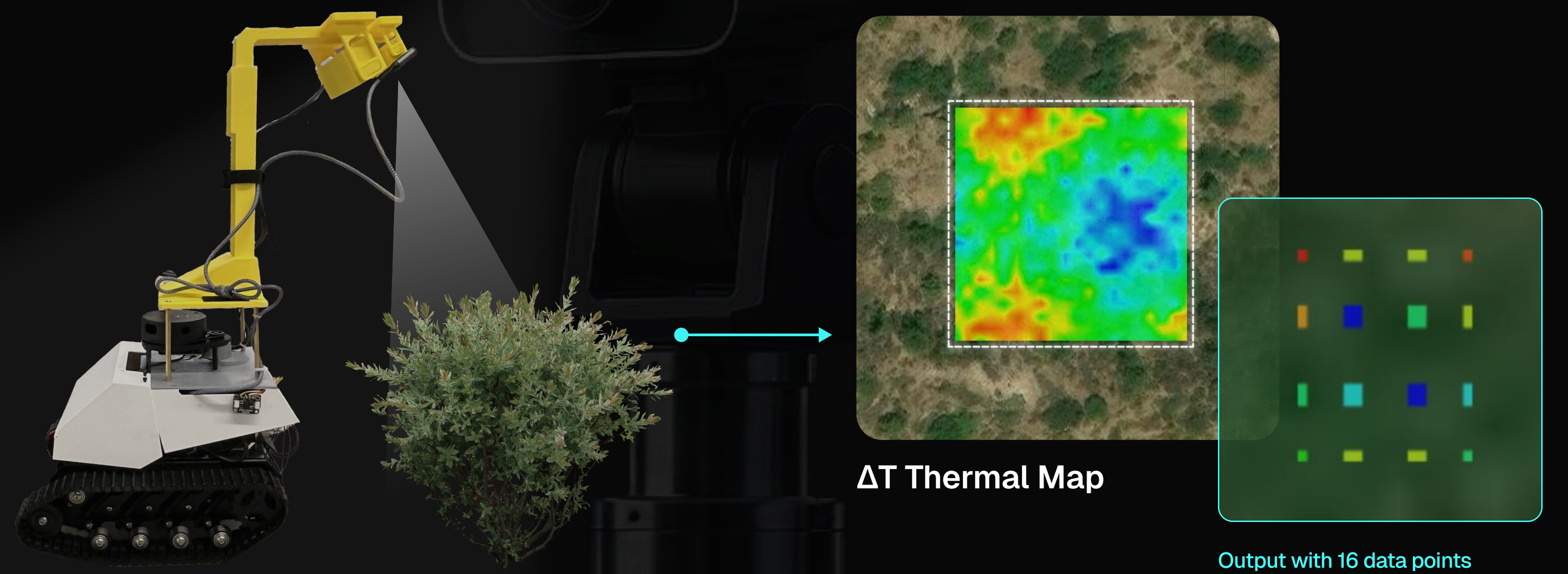
"That was all trees. Oh, you can see the snags that are standing up there. Only it was thicker than that, but they've all been snapped off."

(Describes the severe damage caused by the 2012 Taylor Bridge Fire, which turned 140 acres of forest into a dead wood area)

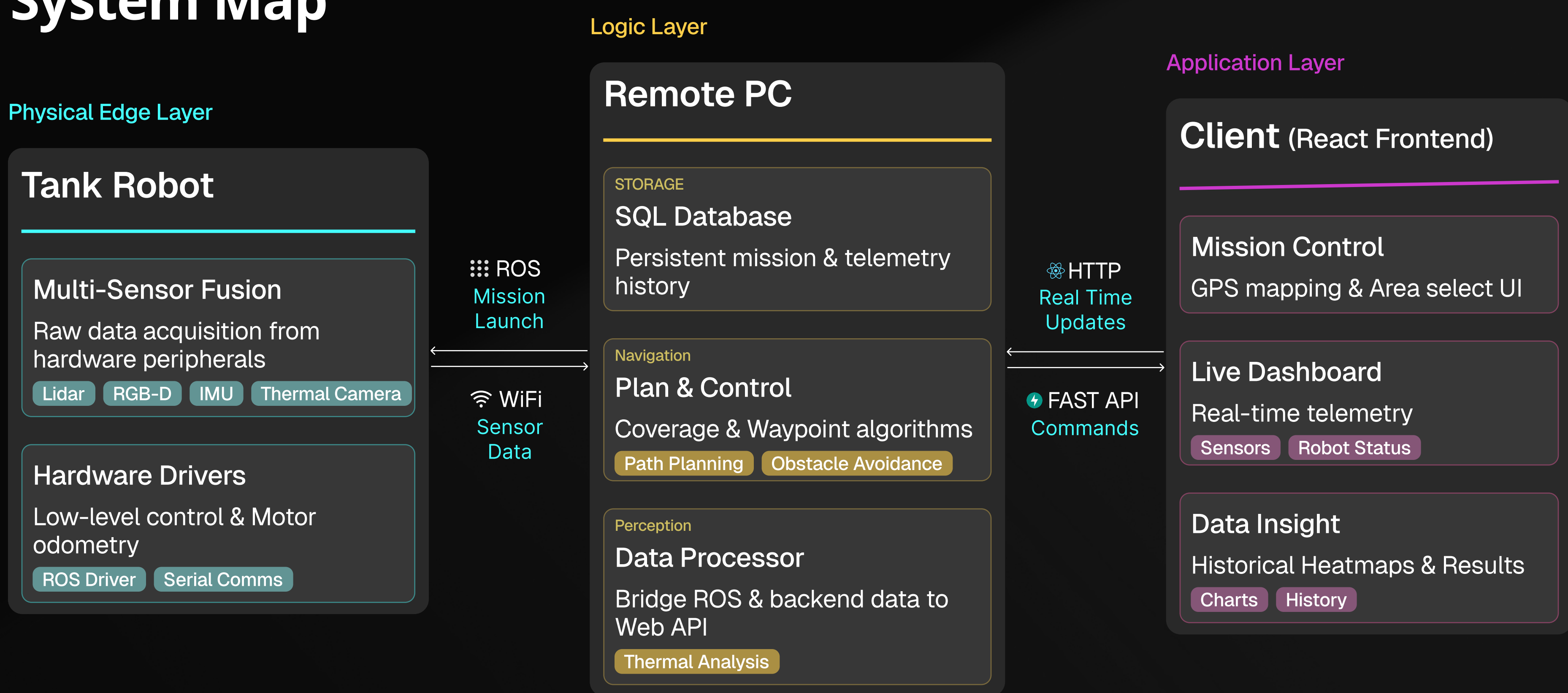


Solution

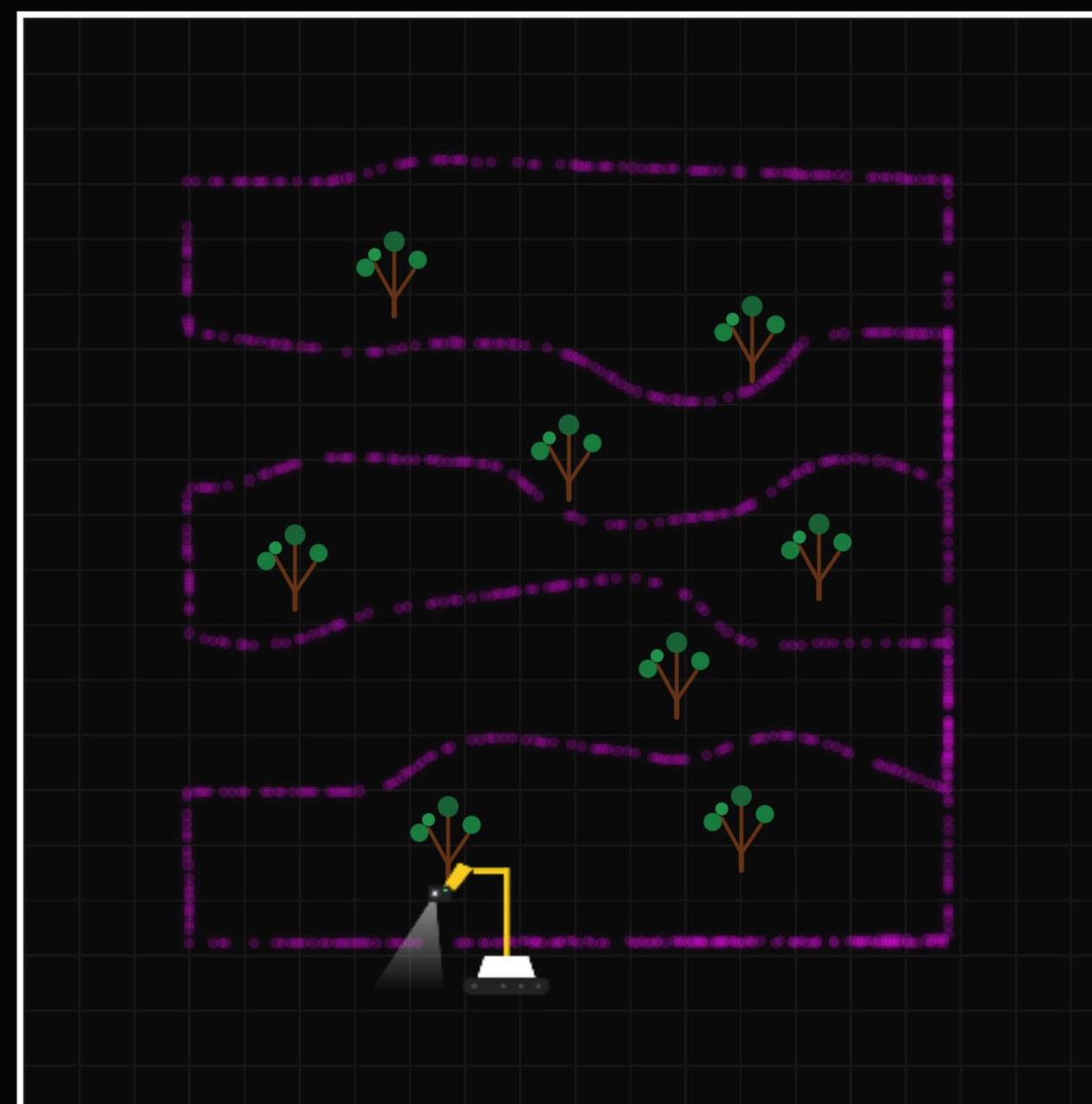
A ground-based autonomous rover system that collects detailed, repeatable multi-modal sensor data in shrubland regions



System Map



Navigation



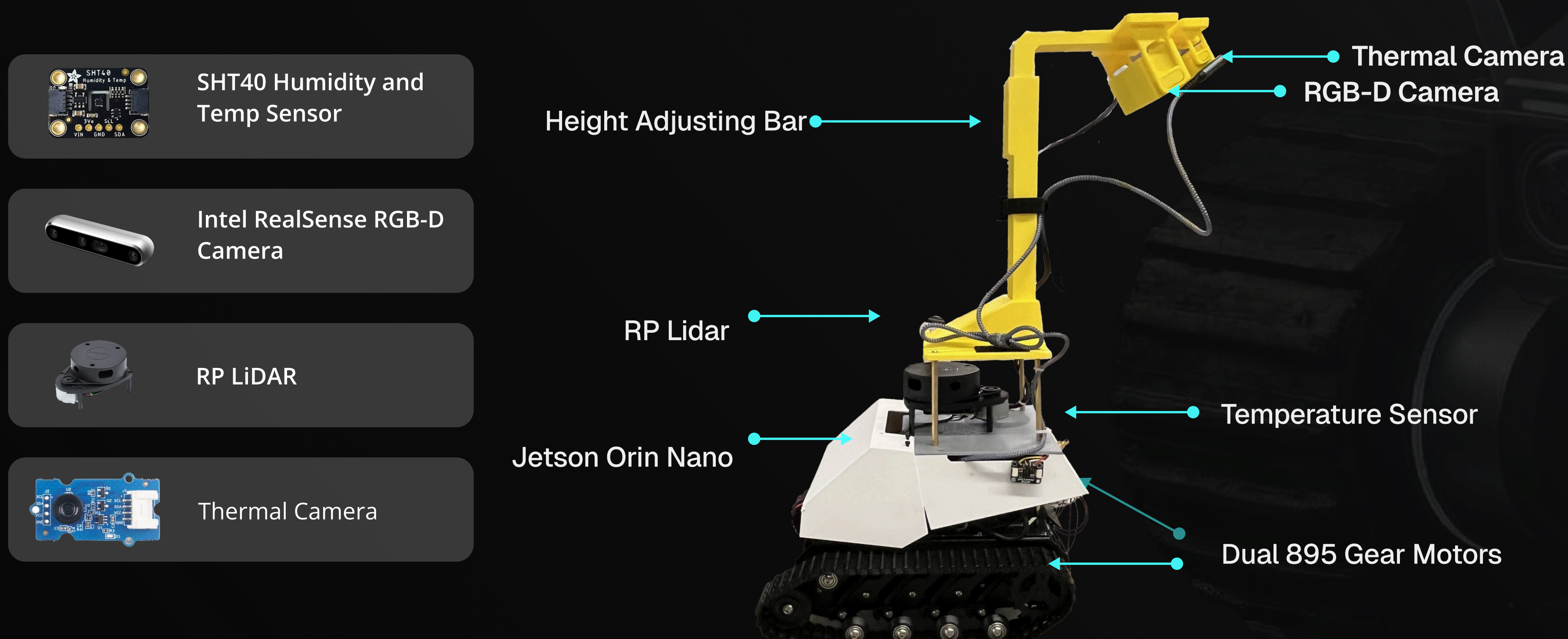
High Coverage Autonomous Navigation

Rover follows a lawnmower coverage path to systematically survey a defined area. Wheel encoder and IMU data are fused with an Extended Kalman Filter (EKF) for reliable localization.

Mapless Obstacle Avoidance

An onboard RPLidar generates a rolling costmap for real-time obstacle avoidance using a Dynamic Window Approach (DWA) planner that enables safe navigation without a pre-built map.

Hardware Components



Dashboard

